ANALYZE E-COMMERCE SALES TO IMPROVE CUSTOMER CONVERSION AND RETENTION

Introduction:

Machine learning projects are not complete when a model is merely trained. A professional workflow also requires clear data preparation, transparent model assumptions, reproducible training steps, appropriate evaluation measures, interpretation of results and proper documentation. The supplied project follows this general workflow by using e-commerce data to test a series of business hypotheses through simple linear regression. The project combines customer, order, order-item, product, review and geolocation data. It creates analytical variables such as Delivery Delay, Fulfillment Time, repeat-purchase indicators, lifetime revenue, loyalty score, average discount, churn and re-engagement. These derived variables are then used to evaluate relationships such
as the effect of delivery performance on customer satisfaction, the relationship between loyalty and lifetime revenue, and the influence of discounts on order value and re-engagement.
The purpose of this report is to convert the project into a structured project document that can be reviewed independently. It therefore explains not only what the code does, but also why each modelling step matters, how the results should be interpreted and where caution is required.

Objectives of the report:

- Describe the datasets and the relationships between them.
- Explain the preprocessing and feature-engineering steps used in the project.
- Document model assumptions, parameters, data splitting and scaling.
- Present and interpret the executed results for H2 through H10.
- Capture evaluation metrics, charts and prediction behaviour transparently.
- Explain the suitability and limitations of the selected modelling approach.
- Provide a final conclusion and recommendations for improving the analysis.

1. Executive Summary:

The project applies simple linear regression as a common validation framework across several business hypotheses. The project loads six related e-commerce datasets, converts relevant fields to usable numeric/date formats, derives delivery and customer-level measures, creates analytical datasets through joins and aggregations, and evaluates hypotheses using train, validation and test partitions.

The modelling function uses a 70% training set, 15% validation set and 15% test set, with random_state=42. Predictor scaling is enabled by default through Standard Scaler. A Linear Regression model is then fitted and evaluated using R^2, MAE and RMSE.

Main findings from the executed outputs

• H7 provides the strongest explanatory performance among the principal results: test $R^2 = 0.2738$, with a positive coefficient of 14,359.0126.

• H8 also shows measurable explanatory power: test $R^2 = 0.1653$ and a positive coefficient of 7,614.5117.

• H2, H3 and H4 have test R^2 values essentially at zero, indicating negligible explanatory power for the respective one-variable linear models.

• H5 has test R^2 □ 0.0001, so the fitted relationship explains almost none of the observed variation in churn.

• H6 has test R^2 □ -0.0002, meaning the model performs no better than a simple baseline in explanatory terms.

• H9 reports a perfect test R^2 of 1.0000 with zero MAE and RMSE. This is an exceptional result and should be treated as a validation warning rather than automatically accepted as evidence of a real-world relationship.

• H10 has a slightly negative test R^2, so the discount-based re-engagement model has negligible explanatory value in its current form.

2. Project Scope and Data Structure:

The project works with six CSV data sources. Their columns indicate a connected e-commerce data model spanning customers, orders, products, order items, reviews and geography.

Dataset	Purpose / Key fields visible in project
customers.csv	Customer identity and segmentation: customer_id, customer_zip_code, gender, age_group, customer_segment.
orders.csv	Order lifecycle and status: order_id, customer_id, order_status, payment_type, purchase/approval/delivery dates, Delivery Delay and Delivery_Status.
order_items.csv	Line-item economics: order_item_id, order_id, quantity, product_id, unit_price, discount(%), shipping_cost, Gross Amount, Discount Amount, Net Product Amount and total cost.
products.csv	Product attributes and economics: product_id, category/subcategory, weight, brand, cost_price, selling_price, stock_availability, Profit and Profit Margin.
order_reviews.csv	Customer feedback: order_id, review_score, review_comment_message and review_date.
geolocation.csv	Geographic enrichment: zip code, city, state, latitude, longitude and region.

Observed order-status distribution

Status	Count	Share of 100,000 orders
delivered	84,971	84.971%
returned	7,000	7.000%
cancelled	5,048	5.048%
shipped	2,980	2.980%

The status counts sum to 100,000, indicating that the executed project is working with a 100,000-order status distribution at this stage.

3. Data Preparation and Feature Engineering:

3.1 Date conversion

The project converts purchase, approval, carrier/delivery and estimated-delivery date columns using pandas.to_datetime with errors='coerce'. This allows invalid or unavailable date values to become missing values rather than stopping execution.

3.2 Delivery Delay

Where the field is not already available, Delivery Delay is calculated as:

Delivery Delay = Actual Customer Delivery Date Estimated Delivery Date

The resulting value is expressed in days. Positive values indicate delivery after the estimated date, while negative values indicate delivery before the estimate.

3.3 Fulfillment Time

Fulfillment Time is derived as:

Fulfillment Time = Customer Delivery Date Purchase Timestamp

3.4 Review preparation

Review scores are converted to numeric values, rows without a usable review score are removed, and review data are joined to order-level delivery measures.

The delivery-delay/review dataset contains 91,971 observations after the inner merge and removal of missing values.

3.5 Customer-level construction

Orders are sorted by customer and purchase timestamp. A purchase_number is assigned using groupby().cumcount()+1. repeat_purchase is then set to 1 when purchase_number is greater than one.

3.6 Order economics

Order-level totals are aggregated from order_items: total_price, total_shipping, mean discount, net product amount and shipping_rate. AOV is represented in the project by the aggregated order amount used in the relevant hypothesis.

3.7 Revenue and loyalty

Customer lifetime revenue is aggregated from order-level values. loyalty_score is defined in the project as total_orders, making loyalty a direct order-count measure.

4. Model Development Methodology:

4.1 Common regression function

The project centralizes model training in a reusable function named simple_linear_regression. It accepts a dataframe, predictor column, target column, hypothesis name and an optional scale flag.

4.2 Cleaning inside the modelling function

• The selected predictor and target are converted to numeric using errors='coerce'.

• Rows with missing predictor or target values are removed.

• The predictor is placed in X and the target in y.

4.3 Data split

Partition	Proportion	Purpose
Training	70%	Fit the regression coefficients.
Validation	15%	Intermediate model assessment.
Test	15%	Final out-of-sample evaluation.

Both train_test_split calls use random_state=42, making the partitioning reproducible.

4.4 Scaling

StandardScaler is fitted on the training predictor and then applied to validation and test data. This prevents information from the validation/test sets being used to estimate the scaling parameters.

4.5 Model and outputs

The estimator is sklearn.linear_model.LinearRegression. The project returns coefficient, intercept, train/validation/test R^2, test MAE and test RMSE, together with the test observations and predictions.

4.6 Evaluation metrics

• R^2 measures the proportion of target variation explained by the fitted model; values near zero indicate little explanatory power and negative test R^2 indicates performance below the mean-prediction baseline.

• MAE measures average absolute prediction error.

• RMSE penalizes larger prediction errors more heavily than MAE.

5. Model Assumptions, Parameters :

5.1 Linear-regression assumptions relevant to this project

- Linearity: the expected relationship between predictor and target should be reasonably linear.
- Independence: observations should not be dependent in a way that invalidates standard regression interpretation.
- Constant error variance: residual spread should be reasonably stable across fitted values.
- Residual behaviour: severe departures from the expected error pattern can reduce the reliability of inference.
- Absence of problematic influential observations: extreme points should be examined because simple regression can be sensitive to them.

The project includes a residual plot for H3, which is an important diagnostic step. However, the project does not execute formal statistical tests for normality, heteroscedasticity, autocorrelation or influence. Therefore, these assumptions are documented as methodological requirements rather than confirmed as satisfied.

5.2 Recorded training parameters

Parameter	**Project setting**
Model	sklearn LinearRegression
Train/validation/test	70% / 15% / 15%
Random seed	42
Scaling	StandardScaler; enabled by default
Missing values	Coerced to NaN and dropped for model inputs
Metrics	Train/validation/test R^2; test MAE; test RMSE

6. Hypothesis Validation: H2 to H4

H2 — Higher product prices improve review scores

Analytical relationship: unit_price review_score. The merged dataset contains 145,585 observations. The coefficient is -0.0028014, test R^2 is -0.0000594, MAE is 1.0458 and RMSE is 1.3364.

The coefficient is slightly negative rather than positive, but the test R^2 is essentially zero. Therefore, the current one-variable linear model provides negligible evidence that product price explains review-score variation.

```
H2 = simple_linear_regression(
    price_review,
    'unit_price',
    'review_score',
    'H2: Higher product prices improve review scores'
)

print("Coefficient:", H2['Coefficient'])
print("Test R²:", H2['Test R²'])
print("Test MAE:", H2['Test MAE'])
print("Test RMSE:", H2['Test RMSE'])
```

```
Coefficient: -0.0028014189975138558
Test R²: -5.935657364242175e-05
Test MAE: 1.0457523685394565
Test RMSE: 1.3363716570800623
```

H3 — Faster/on-time delivery improves customer satisfaction

Analytical relationship: Delivery Delay review_score. The merged dataset contains 91,971 observations. The coefficient is +0.0061498, test R^2 is -0.0001136, MAE is 1.0370 and RMSE is 1.3297.

The fitted coefficient is positive, which is opposite to the directional wording of the hypothesis if positive delay means later delivery. More importantly, the R^2 is effectively zero. The project itself therefore concludes that the relationship has negligible explanatory power.

```
H3 = simple_linear_regression(
    delivery_review,
    'Delivery Delay',
    'review_score',
    'H3: Faster/on-time delivery improves customer satisfaction'
)

print("Coefficient:", H3['Coefficient'])
print("Test R²:", H3['Test R²'])
print("Test MAE:", H3['Test MAE'])
print("Test RMSE:", H3['Test RMSE'])
```

```
Coefficient: 0.006149752515266311
Test R²: -0.00011364104306887945
Test MAE: 1.037024935370729
Test RMSE: 1.3296900782068426
```

H4 — Shorter fulfillment time improves review scores

Analytical relationship: Fulfillment Time review_score. The coefficient is +0.0006746, test R^2 is -0.0002695, MAE is 1.0770 and RMSE is 1.3709.

Again, explanatory power is effectively absent. The small positive coefficient should not be treated as strong business evidence because the model explains essentially none of the variation in review scores.

```
H4 = simple_linear_regression(
    fulfillment_review,
    'Fulfillment Time',
    'review_score',
    'H4: Shorter fulfillment time improves review scores'
)

print("Coefficient:", H4['Coefficient'])
print("Test R²:", H4['Test R²'])
print("Test MAE:", H4['Test MAE'])
print("Test RMSE:", H4['Test RMSE'])
```

```
Coefficient: 0.0006745615350882955
Test R²: -0.0002694939784100292
Test MAE: 1.0769858432786084
Test RMSE: 1.3709018307443384
```

Validation takeaway

H2–H4 demonstrate why coefficient direction alone is not enough. Model quality must be judged using out-of-sample performance and the size of explained variation. For these three hypotheses, the test R^2 values are approximately zero or slightly negative.

7. Hypothesis Validation: H5 to H7

H5 — Delivery delays increase customer churn

Customer churn is defined in the project as total_orders == 1. Average Delivery Delay is calculated at customer level and then merged with the churn indicator.

The coefficient is -0.0028472, test R^2 = 0.0001420, MAE = 0.4988 and RMSE = 0.4994. The negative coefficient does not support the stated positive direction, and the R^2 is essentially zero.

```
H5 = simple_linear_regression(
    customer_churn,
    'Delivery Delay',
    'churn',
    'H5: Delivery delays increase customer churn'
)

print("Coefficient:", H5['Coefficient'])
print("Test R²:", H5['Test R²'])
print("Test MAE:", H5['Test MAE'])
print("Test RMSE:", H5['Test RMSE'])
```

```
Coefficient: -0.002847187051155086
Test R²: 0.00014198568268564138
Test MAE: 0.49880893906088725
Test RMSE: 0.4994452456248702
```

H6 — Higher review scores increase reorder probability

Average review score is calculated per customer, while reorder_rate is defined as 1 when total_orders > 1. The coefficient is -0.0039233, test R^2 = -0.0001993, MAE = 0.4998 and RMSE = 0.4999.

The negative coefficient is opposite to the proposed positive relationship, while the negative R^2 indicates negligible explanatory value.

```
H6 = simple_linear_regression(
    review_reorder,
    'avg_review_score',
    'reorder_rate',
    'H6: Higher review scores increase reorder probability'
)

print("Coefficient:", H6['Coefficient'])
print("Test R²:", H6['Test R²'])
print("Test MAE:", H6['Test MAE'])
print("Test RMSE:", H6['Test RMSE'])
```

```
Coefficient: -0.003923253261701808
Test R²: -0.00019934886205241753
Test MAE: 0.4997661612910785
Test RMSE: 0.4999151584692203
```

H7 — Customer loyalty increases lifetime revenue

Customer lifetime revenue is summed from order-level total_price values. loyalty_score is defined as total_orders. This model produces a coefficient of 14,359.0126 and a test R^2 of 0.2738, with validation R^2 = 0.3032. Test MAE is 12,949.8964 and test RMSE is 22,246.7865.

Among the principal hypotheses evaluated in the results table, H7 has the strongest test R^2. The result indicates a meaningful positive association between the loyalty measure used by the project and lifetime revenue, although 72.62% of target variation remains unexplained by this one predictor.

```
H7 = simple_linear_regression(
    customer_revenue,
    'loyalty_score',
    'lifetime_revenue',
    'H7: Customer loyalty increases lifetime revenue'
)

print("Coefficient:", H7['Coefficient'])
print("Test R²:", H7['Test R²'])
print("Test MAE:", H7['Test MAE'])
print("Test RMSE:", H7['Test RMSE'])
```

```
Coefficient: 14359.012550181878
Test R²: 0.2737654315485979
Test MAE: 12949.896427886442
Test RMSE: 22246.786546601943
```

Business interpretation

Customer order frequency is strongly relevant to customer revenue, but the analysis should not be interpreted as proof that loyalty itself causes higher revenue. In this project, loyalty_score is literally total_orders, while lifetime_revenue is accumulated revenue, so the variables are structurally related.

8. Hypothesis Validation: H8 to H10

H8 — Discount rate influences average order value

The project identifies discount-related fields as discount(%) and Discount Amount. It calculates order-level average discount and combines it with an AOV measure.

The coefficient is +7,614.5117 and test R^2 = 0.1653. Test MAE is 8,702.3882 and RMSE is 15,773.2946.

This is the second-largest R^2 among the reported models. The positive coefficient indicates that higher discount rate is associated with higher AOV in this fitted relationship, but the model still leaves most AOV variation unexplained.

```
H8 = simple_linear_regression(
    discount_aov,
    'discount_rate',
    'AOV',
    'H8: Discount rate influences average order value'
)

print("Coefficient:", H8['Coefficient'])
print("Test R²:", H8['Test R²'])
print("Test MAE:", H8['Test MAE'])
print("Test RMSE:", H8['Test RMSE'])
```

```
Coefficient: 7614.511662567965
Test R²: 0.16533956485867307
Test MAE: 8702.38817384984
Test RMSE: 15773.294643799127
```

H9 — Delivery delays increase cancellation probability

The project converts the order status into a binary cancelled variable and regresses it on Delivery Delay. The reported test R^2 is exactly 1.0000 with MAE and RMSE both 0.0000 and coefficient displayed as -0.0.

This is a highly unusual perfect result. Because the target is binary and the predictor is a delivery-delay variable derived from the order-delivery lifecycle, the result requires careful leakage and data-definition investigation before being treated as a valid business finding.

```
H9 = simple_linear_regression(
    delay_cancel,
    'Delivery Delay',
    'cancelled',
    'H9: Delivery delays increase cancellation probability'
)

print("Coefficient:", H9['Coefficient'])
print("Test R²:", H9['Test R²'])
print("Test MAE:", H9['Test MAE'])
print("Test RMSE:", H9['Test RMSE'])
```

```
Coefficient: -0.0
Test R²: 1.0
Test MAE: 0.0
Test RMSE: 0.0
```

H10 — Discounts increase customer re-engagement

Customer-level average discount is calculated and re-engagement is defined as total_orders > 1. The coefficient is
-0.0033534, test R^2 = -0.0001979, MAE = 0.4836 and RMSE = 0.4914.

The result does not support the proposed positive relationship. Its R^2 is slightly negative, indicating negligible explanatory value.

```
H10 = simple_linear_regression(
    customer_discount,
    'avg_discount',
    'reengagement',
    'H10: Discounts increase customer re-engagement'
)

print("Coefficient:", H10['Coefficient'])
print("Test R²:", H10['Test R²'])
print("Test MAE:", H10['Test MAE'])
print("Test RMSE:", H10['Test RMSE'])
```

```
Coefficient: -0.0033534477166141945
Test R²: -0.00019786928922105318
Test MAE: 0.4835585065686768
Test RMSE: 0.49136331480783957
```

Key caution

H8 is useful as an association signal; H9 is a validation warning; H10 is effectively unsupported by the current linear model. These distinctions should be preserved in any final presentation rather than labeling every hypothesis as accepted or rejected solely from coefficient signs.

9. Consolidated Model Performance

The following table reproduces the principal metrics reported by the project's results table and final evaluation output.

Hyp.	Predictor	Target	n	Coef.	Test R^2	MAE	RMSE
H2	unit_price	review_score	145,585	-0.0028	-0.0001	1.0458	1.3364
H3	Delivery Delay	review_score	91,971	0.0061	-0.0001	1.0370	1.3297
H4	Fulfillment Time	review_score	24,101	0.0007	-0.0003	1.0770	1.3709
H5	Delivery Delay	churn	49,745	-0.0028	0.0001	0.4988	0.4994
H6	avg_review_score	reorder_rate	51,131	-0.0039	-0.0002	0.4998	0.4999
H7	loyalty_score	lifetime_revenue	41,419	14,359.0126	0.2738	12,949.8964	22,246.7865
H8	discount_rate	AOV	—	7,614.5117	0.1653	8,702.3882	15,773.2946
H9	Delivery Delay	cancelled	—	-0.0	1.0000	0.0000	0.0000
H10	avg_discount	reengagement	—	-0.0034	-0.0002	0.4836	0.4914

Performance ranking by test R^2

Rank	Hypothesis	Test R^2	Interpretation
1	H7	0.2738	Strongest among the main executed models; moderate explanatory power.
2	H8	0.1653	Some explanatory power, but substantial variation remains unexplained.
3	H9	1.0000	Perfect result; requires leakage/definition validation.
4	H5	0.0001	Negligible explanatory power.
5	H2	-0.0001	Negligible; effectively no explanatory power.
6	H3	-0.0001	Negligible; effectively no explanatory power.
7	H6	-0.0002	No useful explanatory power.
8	H4	-0.0003	No useful explanatory power.
9	H10	-0.0002	No useful explanatory power.

10. Evaluation Charts and Diagnostic Analysis

The project explicitly stores evaluation charts and prediction outputs. Two H3 charts are included below because H3 is the hypothesis for which the project generated both a fitted-line plot and a residual plot.

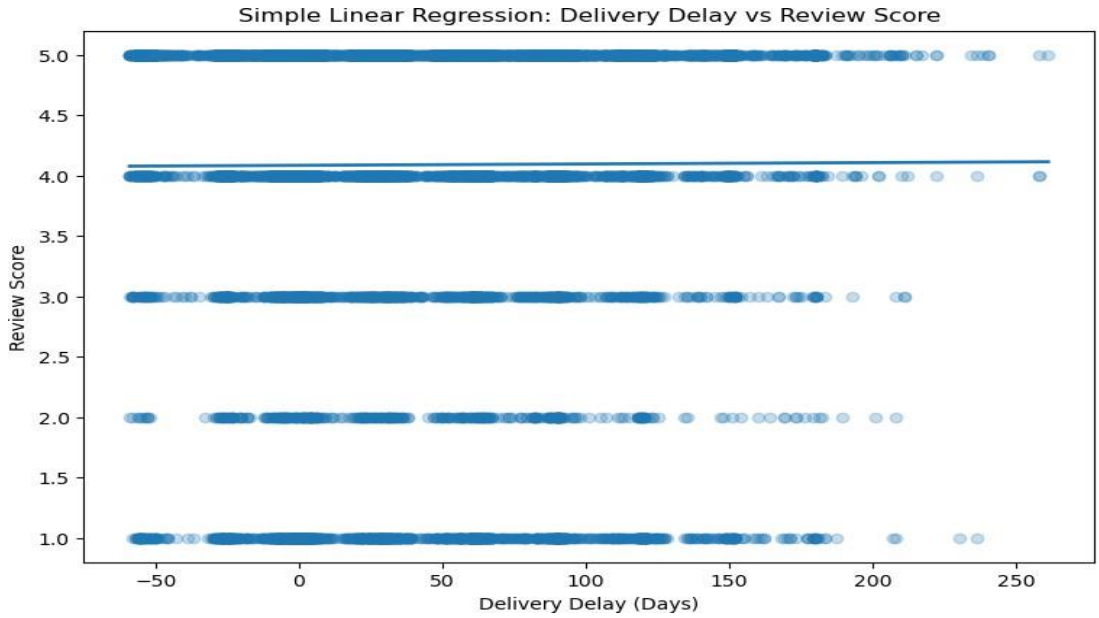


Figure 1. Delivery Delay vs Review Score with fitted regression line. The project generated this chart from H3's test set. The fitted relationship is: **Review Score = 4.0897 + (0.006150 × Delivery Delay)**.

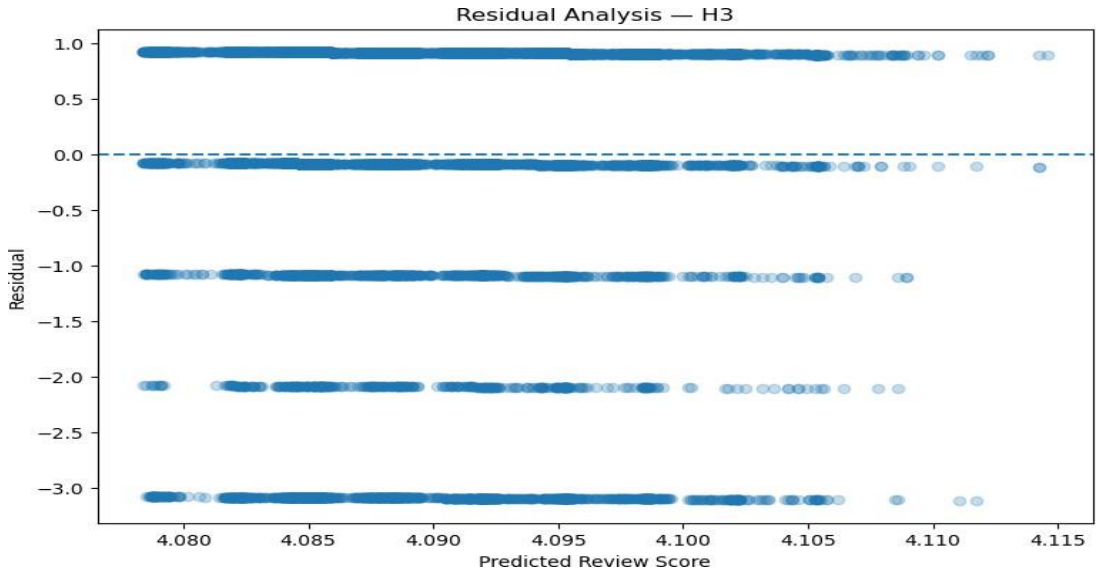


Figure 2. H3 residual analysis. Residuals are calculated as observed review score minus predicted review score and plotted against predicted review score. The project uses a horizontal zero reference line to support visual inspection.

Diagnostic interpretation: the charts help document model behaviour, but a visual diagnostic alone does not prove that all regression assumptions are satisfied. The near-zero test R^2 for H3 remains the dominant performance finding.

11. Final Model Selection, Validation Risks and Limitations

11.1 Model-selection reasoning

The project uses one common model form—simple linear regression—for all hypotheses. This provides consistency and makes the results easy to compare. However, the same model form is not equally suitable for every target variable.

For continuous outcomes such as lifetime revenue and AOV, linear regression is a natural baseline. For binary outcomes such as churn, reorder_rate, cancelled and reengagement, linear regression is generally not the preferred production approach. Logistic regression or other classification models would better represent probability outcomes.

11.2 Important validation risks

• H9's perfect $R^2 = 1.0$ and zero prediction error should be investigated for target leakage, data construction effects or an encoding relationship before acceptance.

• H7 uses total_orders as loyalty_score and also uses total_orders to aggregate lifetime revenue. This creates a structurally linked predictor-target relationship.

• The project defines churn as exactly one order, which is a simplified proxy rather than a time-based churn definition.

• The project uses customer delivery delay averaged across orders for H5; this may mix different order experiences and does not establish temporal causality.

• The analysis is observational. Regression coefficients describe associations in the constructed datasets and do not by themselves establish causation.

• The project suppresses warnings with warnings.filterwarnings('ignore'), which can hide useful modelling or data-quality messages.

11.3 Missing validation steps

• No formal train-vs-validation model-selection procedure is used to tune model complexity; the model remains simple linear regression.

• No confidence intervals or p-values are calculated in the supplied regression function.

• No formal residual normality, heteroscedasticity or influence diagnostics are executed.

• No cross-validation is performed.

These limitations do not invalidate the project as a project workflow, but they limit how strongly the numerical results should be presented as final statistical evidence.

12. Recommendations

Analytical recommendations:

• Retain simple linear regression as a baseline for continuous targets, especially AOV and lifetime revenue.

• For churn, cancellation, reorder and re-engagement, evaluate logistic regression, decision trees, random forests or gradient-boosting classifiers.

• Use multiple predictors rather than one-variable models. Potential predictors visible in the data include category, customer segment, discount, shipping cost, delivery delay, fulfillment time, review score and order history.

• Use cross-validation to estimate generalization more robustly.

• Add confidence intervals and statistical significance tests when the objective is formal hypothesis inference.

• Investigate H9 before reporting it as a successful hypothesis; confirm that the predictor is available before the cancellation outcome and that no target information is embedded in it.

• Refine churn into a time-window definition, such as no purchase within a specified period after the last eligible order, if the business question is genuine customer churn.

13. Conclusion:

The supplied project demonstrates an end-to-end statistical validation workflow for an e-commerce dataset. It loads multiple relational datasets, prepares dates and numeric fields, engineers delivery, fulfillment, customer, revenue and discount measures, builds analytical datasets through merges and aggregations, trains a reproducible linear-regression baseline and evaluates results on held-out test data.

The strongest conventional explanatory results are H7 (loyalty score lifetime revenue, test R^2 0.2738) and H8 (discount rate AOV, test R^2 0.1653). H2, H3, H4, H5, H6 and H10 provide negligible explanatory power under the implemented one-variable regression approach. H9's perfect score is unusual enough that it should be treated as a data/model validation issue until independently verified.

The project therefore succeeds as a documented modelling and validation exercise, while also showing why model choice, target definition, leakage checks and diagnostic testing are essential before turning exploratory regression outputs into business decisions.